

Lab 37 — Data Cleaning with Pandas (Missing Values & Duplicates)

Aim

To identify and handle missing values and duplicate records in a dataset using Pandas.

Algorithm / Procedure

1. Import Pandas and NumPy.
2. Create a DataFrame containing missing values and duplicate records.
3. Identify missing values using `isnull().sum()`.
4. Fill missing values using `fillna()`.
5. Remove duplicate records using `drop_duplicates()`.
6. Display the cleaned DataFrame.

Program

```
import pandas as pd
import numpy as np
df=pd.DataFrame({'name':['A','B','B','C',None],'marks':[78,np.nan,85,92,66]})
print("Missing values per column:\n",df.isnull().sum())
df_filled=df.fillna({'name':'Unknown','marks':df['marks'].mean()})
print("After filling missing values:\n",df_filled)
df_clean=df_filled.drop_duplicates()
print("After removing duplicates:\n",df_clean)
```

Sample Output

Missing values per column:

name 1

marks 1

dtype: int64

After filling missing values:

	name	marks
0	A	78.00
1	B	80.25
2	B	85.00
3	C	92.00
4	Unknown	66.00

After removing duplicates:

	name	marks
0	A	78.00
1	B	80.25
2	B	85.00
3	C	92.00
4	Unknown	66.00

Result

Missing values were successfully identified and filled, and duplicate records were checked and handled using Pandas.